

Task 2

An executable file of veracrypt will be provided to you. Find the address of the entry point of the executable using PE explorer tool and provide the value as the answer as a screenshot.

Attack Name: Portable Executable (PE) File Analysis

Severity: Informational

Reason:

Analyzing the structure of executable files helps security professionals understand how Windows applications are organized and executed. Identifying the Address of Entry Point (AEP) is an important step in reverse engineering and malware analysis, as it indicates where program execution begins.

Steps to Reproduce

Step 1: Launch PE Explorer

PE Explorer was launched, and the provided VeraCrypt Setup 1.26.7.exe file was opened using File → Open.

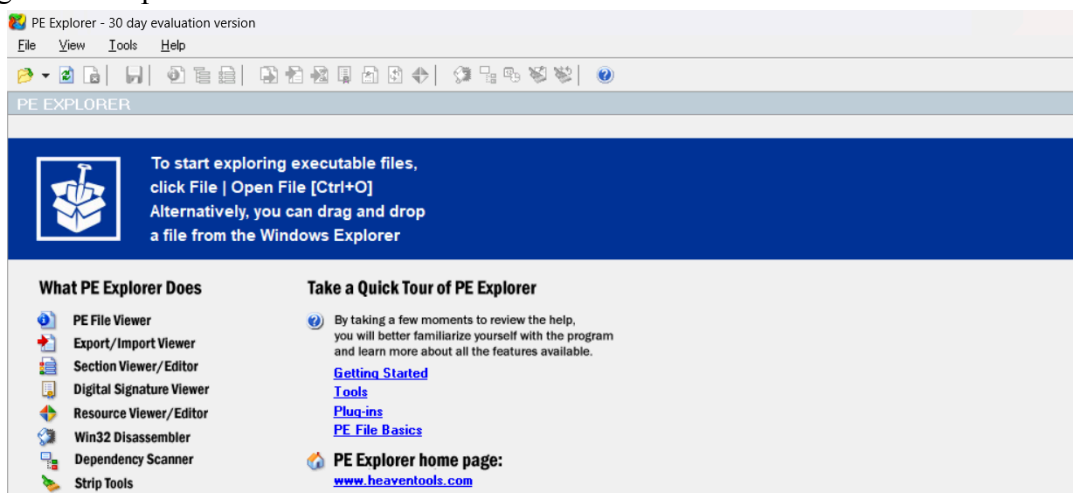
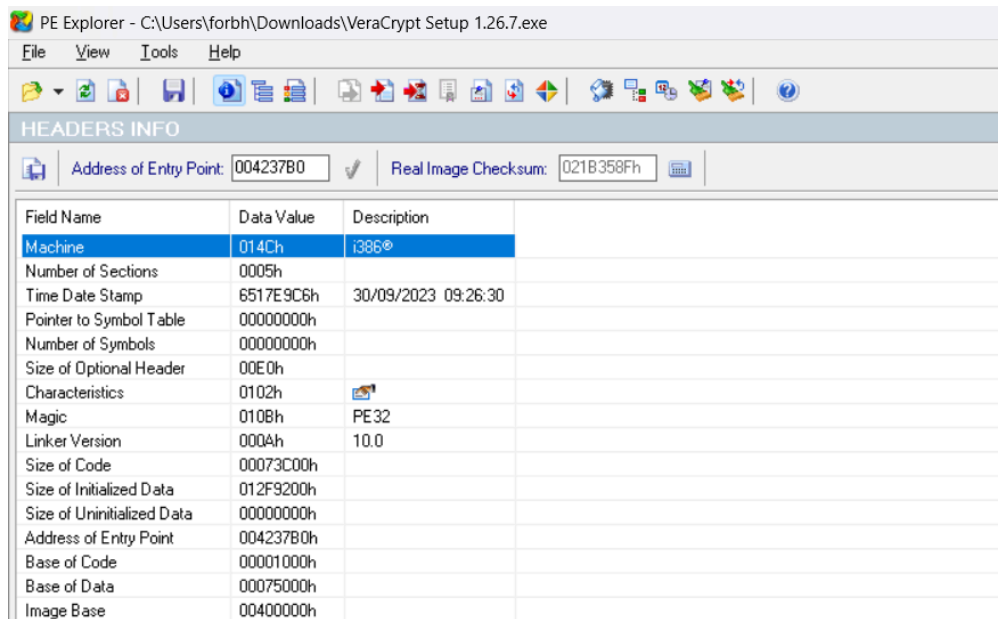


Fig 13: PE explorer interface

Step 2: Locate the Entry Point Address

Navigate to the PE Header section to examine the executable's header information and locate the Address of Entry Point (AEP).



Field Name	Data Value	Description
Machine	014Ch	i386
Number of Sections	0005h	
Time Date Stamp	6517E9C6h	30/09/2023 09:26:30
Pointer to Symbol Table	00000000h	
Number of Symbols	00000000h	
Size of Optional Header	00E0h	
Characteristics	0102h	
Magic	010Bh	PE 32
Linker Version	000Ah	10.0
Size of Code	00073C00h	
Size of Initialized Data	012F9200h	
Size of Uninitialized Data	00000000h	
Address of Entry Point	004237B0h	
Base of Code	00001000h	
Base of Data	00075000h	
Image Base	00400000h	

Fig 14: Entry Point Address

Analysis

The VeraCrypt executable was successfully analyzed using PE Explorer. The tool displayed the internal structure of the Portable Executable (PE) file, including the Address of Entry Point (AEP). The AEP represents the location where the Windows loader transfers control when the executable starts.

Impact

Advantages

1. Helps understand the internal structure of Windows executable files.
2. Assists in reverse engineering and malware analysis.
3. Enables analysts to identify the starting point of program execution for further investigation.

Disadvantages

1. Executable analysis requires specialized tools and technical knowledge.
2. Analyzing untrusted executable files on a production system may pose security risks.
3. Incorrect interpretation of executable information may lead to inaccurate analysis.

Mitigation Steps

1. Analyze executable files only in a secure and isolated environment, such as a virtual machine or sandbox.
2. Verify the authenticity and integrity of executable files before analysis.
3. Keep analysis tools updated to support the latest executable formats.
4. Avoid executing unknown executables directly on the host system.
5. Follow safe reverse engineering practices when examining suspicious files.

Resources Used: PE Explorer, VeraCrypt Setup 1.26.7.exe, Windows Operating System